engineering 4 us all

S Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element F Initial Template

How to use this template
Element F: Consideration of Design Viability
Judgment about Design Viability *(F1)*

Design Viability Description	**Connection to Design Requirement**	**Credible Evidence?**
The solution is cost-effective	The design must not exceed the allocated budget.	A detailed cost breakdown of materials, components, and labor will be provided to ensure costs remain within budget.
The automated watering system functions correctly	The system must water plants at appropriate intervals.	A proof-of-concept test will be conducted to ensure the timer and moisture sensors operate as expected.
The water distribution is efficient and even	The design must ensure that all plants receive adequate water.	A small-scale prototype will be built, and water flow measurements will be taken to confirm even distribution.
The materials are durable and resistant to mold	The materials must not degrade quickly in a wet environment.	Testing different materials for water resistance and mold prevention will be conducted over time.
The system is easy to install and maintain	The design should be user-friendly and require minimal maintenance.	User testing and expert review will assess the ease of installation and long-term maintenance needs.

Name: ______________________________ Date: _________________ Class: _______________

Cost Breakdown for Design Viability

To ensure the design is cost-effective, we will break down the estimated costs of each component:

Component	**Estimated Cost ($)**	**Source / Justification**
Water pump (pressure-based)	$50	Based on market research for small-scale irrigation pumps
PVC pipes and connectors	$30	Pricing from local hardware stores
Automated watering system (timer, moisture sensors)	$80	Prices from online retailers and agricultural suppliers
Water-resistant plant trays	$25	Cost varies based on material (plastic, metal, treated wood)
Antimicrobial coating or mold-resistant material	$15	Research on available coatings for irrigation components
Miscellaneous (wiring, mounting brackets, fasteners)	$20	Estimated based on required hardware
Total Estimated Cost	$220	Ensures feasibility within a reasonable budget

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Testing Procedures for Validation

To confirm the functionality and viability of our design, we will perform several tests:

1. Automated Watering System Test

- **Objective:** Ensure the timer and moisture sensors correctly regulate watering intervals.
- **Procedure:**
    1. Set up the system with a small plant bed.
    2. Program the timer for different watering schedules.
    3. Measure the soil moisture before and after each watering cycle.
    4. Adjust settings if water levels are too high or too low.
- **Success Criteria:** System consistently waters plants at the appropriate intervals without manual intervention.

2. Water Distribution Efficiency Test

- **Objective:** Verify that all plants receive an equal amount of water.
- **Procedure:**
    1. Place measuring cups at different points under the watering system.
    2. Run the system for a set time and record the amount of water collected in each cup.
    3. Adjust pipe angles or nozzle types if distribution is uneven.
- **Success Criteria:** Water is evenly distributed across all plant areas.

3. Mold and Durability Test

- **Objective:** Determine if the materials are resistant to mold growth and degradation.
- **Procedure:**
    1. Expose different materials (plastic, treated wood, metal) to constant moisture for two weeks.
    2. Observe any mold formation, warping, or degradation.
    3. Compare results to select the most durable materials.
- **Success Criteria:** Selected materials show no significant mold growth or structural damage.

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4. User-Friendliness Test

- **Objective:** Ensure the system is easy to install and maintain.
- **Procedure:**
    1. Have a small group of users attempt to set up and operate the system with minimal instructions.
    2. Collect feedback on difficulty level and suggested improvements.
- **Success Criteria:** Users can install and operate the system with little or no external help.